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Association Between Biofilm Formation and Extended-Spectrum Beta-Lactamase Production in *Klebsiella pneumoniae* Isolated from Fresh Fruits and Vegetables

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Abstract
The presence of extended-spectrum beta-lactamase (ESBL)-producing *Klebsiella pneumoniae* in fresh fruits and vegetables is a growing public health concern. The primary objective of this study was to investigate the relationship between biofilm formation and extended-spectrum β-lactamase (ESBL) production in *K. pneumoniae* strains obtained from fresh fruits and vegetables. Out of 120 samples analysed, 94 samples (78%) were found to be positive for *K. pneumoniae*. Among the *K. pneumoniae* strains isolated, 74.5% were from vegetables, whereas the remaining (25.5%) were from fresh fruits. *K. pneumoniae* isolates were resistant to at least three different classes of antibiotics, with ceftazidime (90%) and cefotaxime (70%) showing the highest resistance rates. While the high occurrence of ESBL-producing and biofilm-forming *K. pneumoniae* strains were detected in vegetables (73.5% and 73.7%, respectively), considerable amounts of the same were also found in fresh fruits (26.5% and 26.3%, respectively). The results further showed a statistically significant ($P < 0.001$) association between biofilm formation and ESBL production in *K. pneumoniae* strains isolated from fresh fruits and vegetables. Furthermore, the majority (81%) of the ESBL-producing strains harbored the $bla_{CTX\text{-}M}$ gene, while a smaller proportion of strains carried the bla_{TEM} gene (30%), bla_{SHV} gene (11%) or bla_{OXA} (8%). This study highlights the potential public health threat posed by *K. pneumoniae* in fresh fruits and vegetables and emphasizes the need for strict surveillance and control measures.

Introduction

Klebsiella pneumoniae is one of the Gram-negative, rod-shaped bacteria involved in hospital-acquired infections, including pneumonia, urinary tract infections, and bloodstream infections [1]. Recently, there has been a notable increase in the occurrence of infections attributed to multidrug-resistant (MDR) strains of *K. pneumoniae*, with a specific emphasis on strains that produce extended-spectrum β-lactamases (ESBLs) [2]. ESBLs are enzymes that confer resistance to a broad range of beta-lactam antibiotics, including penicillins, cephalosporins, and monobactams. ESBL-producing *K. pneumoniae* strains are resistant to several classes of antibiotics, making infections difficult to treat and hence increasing the risk of morbidity and mortality [3, 4]. The prevalence of ESBL-producing *K. pneumoniae* strains has surged globally, emerging as a notable concern for society at large. These bacteria pose a significant threat beyond healthcare settings, with the potential to trigger outbreaks and spread swiftly within communities. Moreover, there is growing apprehension about their ability to infiltrate various facets of society and the environment, raising broader public health implications [2].

Foodborne infections caused by *K. pneumoniae* are of growing concern due to the bacterium's ability to survive in food environments and its potential to infect humans through the consumption of contaminated foods [5]. Fresh fruits and vegetables have been implicated as potential reservoirs for *K. pneumoniae*, and recent studies have suggested that these food sources may be contaminated with ESBL-producing strains [6]. Biofilm formation is a well-known virulence factor in *K. pneumoniae* [7] and contributes to the persistence of the bacterium in various environments [8, 9]. Recent

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studies have shown that *K. pneumoniae* has the capability to form biofilms on food surfaces [10, 11]. A study conducted in India found that 25.42% of ready-to-eat street foods were positive for ESBL-producing *K. pneumoniae* [12]. Studies have also investigated the prevalence of biofilm formation in *K. pneumoniae* isolated from food environments. In a study conducted in Norway, 97.5% of *K. pneumoniae* isolates from broilers and turkeys were able to form biofilms [13]. Furthermore, food fishes showed a high prevalence of multidrug-resistant (MDR) strains of *K. pneumoniae* with a highest rate of ESBLs and virulence genes [14]. Moreover, there is growing evidence of *K. pneumoniae* role in community-acquired infections, particularly in the context of foodborne transmission [15]. Antimicrobial resistance poses substantial economic challenges at both the individual and societal levels. However, the relationship between biofilm formation and ESBL production in *K. pneumoniae* isolated from fresh fruits and vegetables remains largely unexplored. Investigating the presence of ESBL-producing *K. pneumoniae* in fresh produce is essential to understand the potential reservoir of antimicrobial resistance in the community.

We hypothesize that there is a positive association between the ability of *K. pneumoniae* to form biofilms and its tendency to produce ESBLs, suggesting a potential reservoir of antimicrobial resistance in fresh produce. The outcomes of this investigation are anticipated to yield valuable insights into the epidemiology of ESBL-producing *K. pneumoniae* in food environments, thereby facilitating the formulation of strategies to reduce the risk of foodborne infections caused by these MDR pathogens.

Materials and Methods

Sample Collection and Identification

Fresh fruit and vegetable samples were collected from different local markets in the city. A total of 120 samples were collected from the local market of Mangalore, India, including 53 samples of fruits (papaya, guava, and banana) and 67 samples of vegetables (coriander, spinach, cabbage, and tomato) (random sampling method was used). The collected samples were placed in an icebox and processed within 2 h of collection. Sterile distilled water was used to clean the surface, and 10 g of each sample were weighed and homogenized in a sterile mortar and pestle. The resulting homogenized mixture was then transferred to 0.1% buffered peptone water (90 ml) and incubated at 37 °C for 24 h. The enriched sample was serially diluted (1:10 dilution) and plated on ESBL-selective agar (Himedia, India) and incubated overnight at 37 °C. Blue-green colonies and morphologically distinct colonies of *K. pneumoniae* were isolated from each plate with selective media and streaked onto nutrient agar [16].

Identification of *K. pneumoniae*

To confirm the identity of *K. pneumoniae* and to further validate, the representative strains from each sample were identified as per the 16S rRNA gene amplification protocol [17]. Briefly, DNA extraction from these isolates followed the manufacturer's protocol using the QIAamp DNA minikit from Qiagen, Bangalore. Subsequently, a PCR analysis was carried out on these isolates in a 25 µl reaction volume to discern the strains. Each PCR reaction comprised 1 µl of both forward and reverse primers, 12 µl of PCR mastermix, 1 µl of template, and 10 µl of nuclease-free water for amplification of the 16S rRNA gene, utilizing universal primers (27F and 1429R, Table 1). The PCR product was analysed through agarose gel electrophoresis, purified and sequenced (~ 1500 bp). The BLAST and NCBI databases were used to identify the bacterial species, and the obtained sequences were deposited at GenBank.

Antibiotic Susceptibility Testing

Antibiotic susceptibility testing was performed using the disk diffusion method [13] as per the guidelines of the

Table 1 Primers used for 16S rRNA gene sequencing and detection of gene encoding ESBLs in this study

Primer name	Primer sequence (5′→3′)	Target gene	Reference
27F	AGAGTTTGATCCTGGCTCAG	16S rRNA	[35]
1492R	TACGGYTACCTTGTTACGACTT		
SHV-F	AGGATTGACTGCCTTTTTG	bal_{SHV}	[36]
SHV-R	ATTTGCTGATTTCGCTCG		
TEM- F	TTGGGTGCACGAGTGGGTTA	bal_{TEM}	[37]
TEM-R	TAATTGTTGCCGGGAAGCTA		
CTX M II- F	ACCGCCGATAATTCGCAGAT	$bal_{CTX\text{-}M\ II}$	[19, 38]
CTX M II -R	GATATCGTTGGTGGTGCCATAA		
OXA-1 F	ATGAAAAACACAATACATATCAACTTCGC	$bal_{OXA\text{-}1}$	[39]
OXA-1 R	GTGTGTTTAGAATGGTGATCGCATT		

clinical and laboratory standards institute (CLSI, 2020). The tested antibiotics included amoxicillin-clavulanic acid (30 µg), cefotaxime (30 µg), ceftazidime (30 µg), cefepime (30 µg), imipenem (10 µg), meropenem (10 µg), ciprofloxacin (5 µg), and gentamicin (10 µg). The results were interpreted according to the CLSI guidelines [18].

Biofilm Formation Assay

Crystal Violet Staining

Biofilm formation was assessed using the microtiter plate method with some modifications [19]. Briefly, a single colony of strains was inoculated into 5 ml of tryptic soy broth (TSB) and incubated at 37 °C overnight with shaking at 200 rpm. 100 µl of the bacterial suspension ($1X10^8$ CFU/ml) was added to each well of a sterile flat-bottomed 96-well polystyrene microtiter plate. The plate was incubated at 37 °C for 24 h without shaking. After incubation, the wells were washed three times with phosphate-buffered saline (PBS) to remove non-adherent bacteria, and the biofilms were fixed with 100 µl of 99% methanol for 15 min. The methanol was then removed, and the wells were air dried. The biofilms were stained with 100 µl of 0.1% (w/v) crystal violet for 10 min. After staining, the excess crystal violet was removed, and the wells were washed three times with distilled water. The bound crystal violet was solubilized with 200 µl of 95% ethanol for 30 min, and the optical density (OD) was measured at 570 nm using a microtiter plate reader. Biofilm production was evaluated based on established criteria [20, 21], and the bacteria were classified according to their biofilm-forming capabilities, categorized as either weak ($OD \leq 0.74$), moderate ($0.74 < OD \leq 1.48$), or strong ($OD > 1.48$) biofilm producers.

Fluorescence Microscopy

Fluorescence microscopy was employed for the analysis of the total biofilm biomass, following the previously described methodology [22]. Bacterial cells that had been cultured overnight (at a concentration of $1.5X10^8$ CFU/ml) were inoculated into a petri plate containing a slide positioned at the bottom surface, along with LB broth supplemented with 10% glucose. The petri plate was then incubated at 37 °C for 24 and 48 h. Subsequently, the slides were washed with PBS buffer and fixed using 1% methanol. To facilitate staining, the slides were treated with 0.01% acridine orange and incubated at 28 °C for 2 min. After washing the slides with sterile water, they were air dried and examined under a fluorescence microscope (Euromax; excitation at 490 nm; emission at 525 nm). Moreover, the fluorescence intensity of the images ($n = 6$) was quantified using Fiji software and subsequently compared to evaluate the biomass observed at both 24 h and 48 h of incubation.

Extended-Spectrum *Beta*-Lactamase (ESBL) Production Assay

E-Tests for ESBL Detection

E-test strips were utilized to detect ESBLs in *K. pneumoniae* isolates [23]. These strips are phenotypic ESBL detection tools that are coated with two different concentrations of ceftazidime: one containing clavulanic acid (CAZ + : 0.016–1 µg/ml) and the other without clavulanic acid (CAZ: 0.25–16 µg/ml). Both concentrations were placed on a single strip in a gradient pattern (Himedia, Mumbai). The E-test procedure, along with reading and interpretation, was conducted following the manufacturer's guidelines. To initiate the test, the strains from an overnight plate were suspended in a saline solution (0.85% NaCl) to achieve an inoculum with a concentration equivalent to $1X10^8$ CFU/ml. This suspension was then spread on a Mueller–Hinton agar plate, and an ESBL E-test strip was applied to the agar's surface. Subsequently, the plate was incubated at 37 °C overnight. Positive ESBL results were identified when the ratio of the value obtained for ceftazidime (CAZ) to the value of ceftazidime in combination with clavulanic acid (CAZ +) was greater than 8.

Phenotypic Confirmation of ESBL

Phenotypic confirmation of ESBL was performed as per previously reported methods [19]. Briefly, a bacterial suspension of *K. pneumoniae* was spread onto Mueller–Hinton agar plates, and discs containing ceftazidime (CAZ), ceftazidime with clavulanic acid (CAZ-CA) and aztreonam (AZ) were placed 20 mm apart on the surface of the agar plate. The plate was incubated at 37 °C for 24 h, and the inhibition zones around the disks were measured. A positive indication for ESBL production was identified when the growth-inhibitory zone diameter of CAZ-CA discs exhibited an increase of at least 5 mm compared to the zone diameter of CAZ discs following incubation (CLSI, 2019).

ESBL Gene Amplification

The ESBL genes were amplified using the polymerase chain reaction (PCR) method [24]. Genomic DNA was extracted from each *K. pneumoniae* isolate using a commercial DNA extraction kit following the manufacturer's instructions (Qiagen, Bangalore). The extracted DNA was then used as a template for PCR amplification of the ESBL genes using specific primers targeting the $bla_{CTX\text{-}M\ II}$, bla_{SHV}, bla_{OXA} and bla_{TEM} genes (Table 1). PCR reactions were carried out in a thermal cycler using the following conditions: an initial denaturation step at 95 °C for 5 min, followed by 30 cycles of denaturation at 95 °C for 30 s, annealing at the specific temperature for each primer set (55 °C for $bla_{CTX\text{-}M\ II}$ and bla_{OXA}, 56 °C for bla_{SHV} and bla_{TEM}) for 30 s, and extension at 72 °C for 30 s. A final extension step at 72 °C for 5 min was performed to complete the reaction. PCR products were analyzed by gel electrophoresis on a 1.5% agarose stained with ethidium bromide. PCR products were considered positive for ESBL genes if an amplicon of the expected size of the amplified fragment was observed. Furthermore, a positive control was used to check the appropriate amplification of ESBL gene from a ESBL-producing *E. coli* (lab collection). Negative controls (without template DNA) were included in each run to validate the PCR results. Additionally, in the multiplexed reaction, the process involved an initial denaturation at 98 °C for 45 s, annealing at 50 °C for 45 s, and elongation at 72 °C for 1 min. The resulting PCR products were then analysed by electrophoresis to detect the presence of antimicrobial resistant genes.

Statistical Analysis

The data obtained were analysed using the GraphPad Prim software. The chi-square test was used to determine the association between biofilm formation and ESBL production. The biofilm development ability was analysed by one way ANOVA followed by Tukey's test. A P-value of < 0.05 was considered statistically significant.

Results

Prevalence of *K. pneumoniae*

A total of 120 samples, including 53 fresh fruits and 67 vegetables, were collected from different local markets in the study area (Mangalore, India). Out of the 120 total samples analysed, a substantial number of isolates (94) were found to be *K. pneumoniae*. Among these isolates, 70 isolates were derived from vegetables, whereas 24 isolates were obtained from fresh fruits (Table 2 and Table S3). Out of 6 papaya samples analysed, 2 isolates were found to be ESBL-positive *K. pneumoniae*. Among the 21 guava samples analysed, all 12 isolates were found to be ESBL-positive. Similarly, 26 banana samples analysed, 10 strains of *K. pneumoniae* were isolated in which 8 strains displayed ESBL-positivity. Among the 20 coriander samples analysed, 16 isolates were found to be *K. pneumoniae*, in which 12 were ESBL-positive. Spinach, cabbage, and tomato samples also displayed varying proportions of ESBL positivity. Overall, 83 *K. pneumoniae* isolates were ESBL-positive, highlighting potential concerns regarding antimicrobial resistance in these food samples.

Antibiotic Susceptibility

Antibiotic susceptibility testing on *K. pneumoniae* isolates was performed using the disc diffusion method. The results showed that all the isolates were resistant to at least three different classes of antibiotics (Table S1). In particular, the highest resistance was observed against ceftazidime (90%), followed by cefotaxime (70%). The isolates showed relatively lower resistance to imipenem (4%) and meropenem (12%). The high levels of resistance observed against commonly used antibiotics such as ceftazidime and cefotaxime signify the limited treatment options available for infections caused by these MDR *K. pneumoniae* strains. However, it is noteworthy that the isolates exhibited relatively lower resistance rates against imipenem and meropenem, which are carbapenem antibiotics considered to be critical therapeutic options for MDR infections.

Biofilm Formation Assay

Biofilm formation was assessed by the microtiter plate method (Fig. 1). Out of the 94 *K. pneumoniae* isolates, 76 strains formed the biofilm. The strains were further

Table 2 Sample-wise distribution of *Klebsiella pneumoniae*, and corresponding ESBL$^+$ and the biofilm$^+$ isolates

Sl. No	Sample name	Sample count	Total count of *K. pneumoniae* strains	Count of ESBL$^+$ strains	Count of biofilm$^+$ strains	Count of biofilm$^+$ and ESBL$^+$ strains
1.	Papaya	6	2	2	20	20
2.	Guava	21	12	12		
3.	Banana	26	10	8		
4.	Coriander	20	16	12	56	54
5.	Spinach	15	15	14		
6.	Cabbage	10	27	24		
7.	Tomato	22	12	11		
Total		120	94	83	76	74

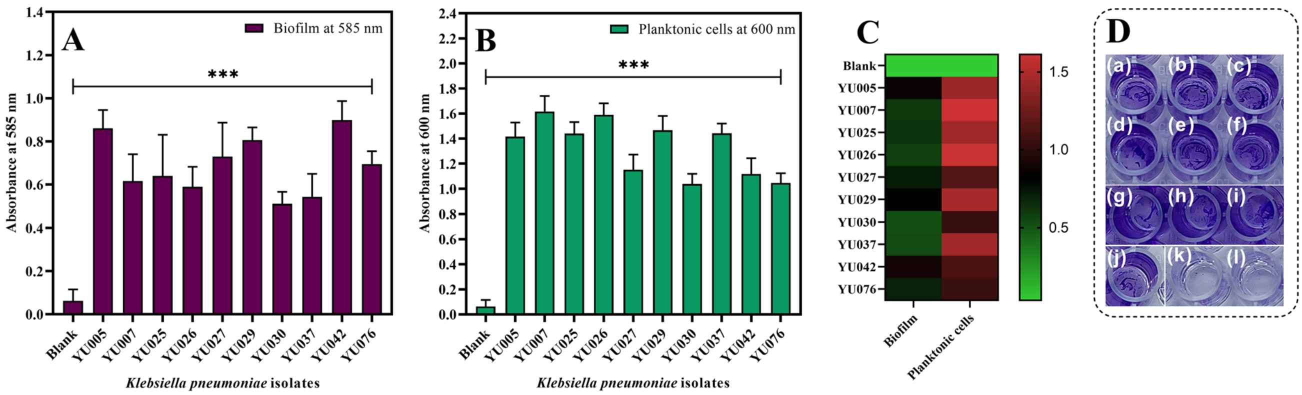

Fig. 1 The biofilm formation ability of the representative *K. pneumoniae* strains as determined on polystyrene surface. The biofilms formed after 24 h of incubation were analyzed and quantified **a**. Additionally, planktonic cells were quantified **b**. Heatmap shows differential biofilm forming ability of *K. pneumoniae* isolates **c**. Biofilm formation was assessed on a 96-well microtiter plate by the strains *K. pneumoniae* **d**, YU005 to YU076 and blank has shown in the picture as (a) to (j) and (k) to (l), respectively. Ossne-way ANOVA followed by Tukey's test was used to test the significance between the groups. ***$P < 0.001$; Error bar, mean ± SD ($n = 3$)

classified as weak (28%), moderate (28%), and strong (44%) biofilm producers. Among the 76 biofilm-positive strains, the majority (56) were from vegetables, while the remaining 20 were from fresh fruits (Table 2). Interestingly, 18 ESBL-positive strains failed to form biofilm. Additionally, the fluorescence analysis of the formed biofilms revealed variations in biomass development among the isolates at 37 °C ($P < 0.001$) (Fig. 2a). Furthermore, a significant number of planktonic cells were quantified (Fig. 1b), indicating presence of bacterial cells suspended in the liquid medium. These data indicated differential biofilm-forming abilities of *K. pneumoniae* isolates originated from the tested samples under the specified temperature conditions.

Phylogenetic Analysis and ESBLs Test

The representative biofilm-forming isolates were identified through 16S rRNA gene sequencing and subjected to phylogenetic analysis with the input of closely related reference strains from clinical and environmental samples. The resulting phylogram (Fig. 3a) exhibited a distinct subclade consisting of the isolated *K. pneumoniae* strains, which demonstrated a close relationship to pathogenic or clinical strains. This observation strongly suggests that the isolates possess a greater prevalence of pathogenicity or virulence-related characteristics.

E-strips were employed to detect ESBLs in all 94 strains of *K. pneumoniae* obtained from tested samples. The results from the E-strips revealed that a significant majority (88.8%) of *K. pneumoniae* isolates were ESBL producers (Table S2). This finding indicates a high prevalence of

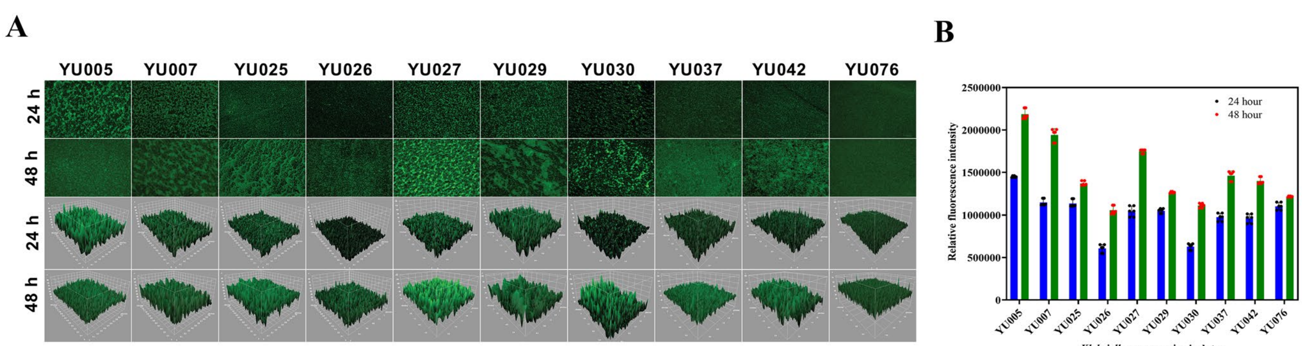

Fig. 2 Characterization of biofilms of the representative *K. pneumoniae* strains using fluorescence microscopy (**a**) in 2D (upper panels) and 3D (lower panels). Cells of different strains of *K. pneumoniae* grown on glass slides were subjected to fluorescence microscopy after staining with acridine orange at 24 h and 48 h. The mean fluorescence intensity of the biofilm formation of different *K. pneumoniae* strains were determined using the Fiji software (**b**)

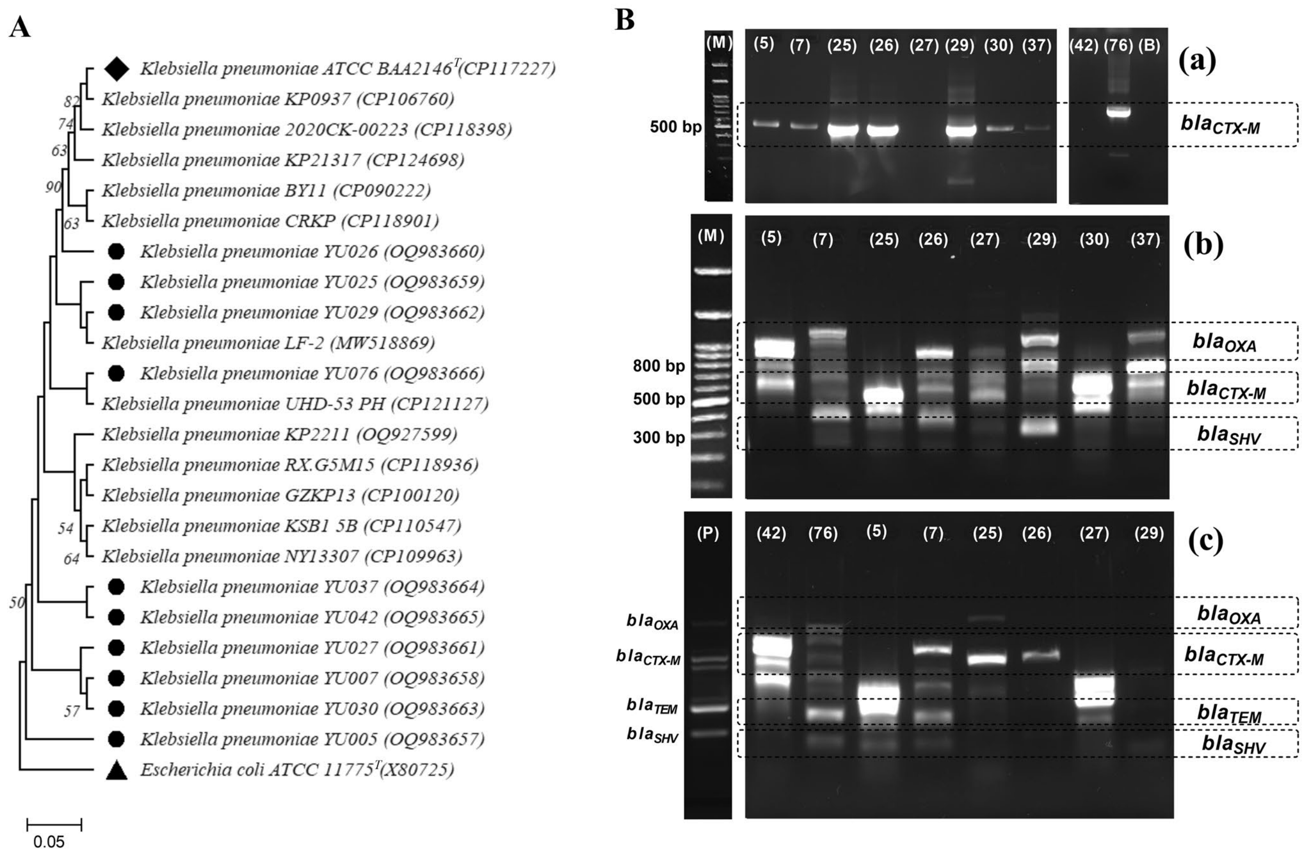

Fig. 3 Phylogenetic analysis of the representative *K. pneumoniae* strains that produce ESBLs. **a** A neighbor-joining phylogenetic tree was constructed to illustrate the evolutionary relationship between *K. pneumoniae* strains and their closest BLAST hits. The tree was generated using the MEGA-11 software and supported by 1000 bootstraps, ensuring the robustness of the tree topology. **b** A multiplex PCR method was employed to identify the presence of four specific beta-lactamase genes, namely $bla_{CTX\text{-}M\ II}$, bla_{SHV}, bla_{OXA} and bla_{TEM} in *K. pneumoniae* strains that were isolated from fresh fruits and vegetables. *Lane 5 to 76 K. pneumoniae* strains, *Lane B* Blank, without DNA template, *Lane M* Marker DNA and *Lane P* Positive control (a, b and c)

ESBL-producing *K. pneumoniae* in the samples obtained from fresh fruits and vegetables. The results also showed that 97.4% of biofilm-forming *K. pneumoniae* strains were found to be ESBL producers. ESBL production is a significant mechanism of antibiotic resistance in Gram-negative bacteria, including *K. pneumoniae*. The study showed interesting findings regarding the association between biofilm formation and ESBL production in *K. pneumoniae* strains isolated from fresh fruits and vegetables. The statistical analysis, specifically using the chi-square test, demonstrated a highly significant association between these two traits ($P < 0.001$). Remarkably, the proportion of *K. pneumoniae* strains that exhibited biofilm formation and also produced ESBLs was significantly higher compared to the proportion of strains that did not form biofilms (Table 2). These results highlight the potential interplay between biofilm formation and the production of ESBLs in *K. pneumoniae* strains isolated from fresh fruits and vegetables.

In addition to examining biofilm formation and ESBL production, the study also investigated the genetic characteristics of the isolated *K. pneumoniae* strains using multiplexed PCR assays. The analysis provided insights into the presence of specific resistance genes associated with ESBL production in these strains. The results revealed that the majority of the ESBL-producing strains (81%) carried the $bla_{CTX\text{-}M}$ gene (Table S3). This finding aligns with previous studies that have consistently reported the prevalence of the $bla_{CTX\text{-}M}$ gene among ESBL-producing *K. pneumoniae* strains. The $bla_{CTX\text{-}M}$ gene confers resistance to extended-spectrum cephalosporins, which is consistent with the observed ESBL phenotype in these strains. In addition to the $bla_{CTX\text{-}M}$ gene, a smaller proportion of strains were found to harbor other resistance genes. Specifically, 30% of the ESBL-producing strains carried the bla_{TEM} gene. The bla_{SHV} gene was detected in 11% of the strains, and the bla_{OXA} gene was found in 8% of the strains (Fig. 3b). These genes are also known to be associated with resistance to beta-lactam antibiotics. These results provide valuable genetic insights into the mechanisms underlying ESBL production in *K. pneumoniae* strains isolated from fresh fruits and vegetables. The predominance of the $bla_{CTX\text{-}M}$ gene, along with the presence of other resistance genes, highlights the diverse genetic backgrounds and resistance mechanisms employed by these MDR pathogens.

Discussion

The present study aimed to investigate the association between biofilm formation and ESBL production in *K. pneumoniae* isolated from fresh fruits and vegetables. High occurrence (74.5%) of *K. pneumoniae* in vegetables, followed by fresh fruits (25.5%) emphasises the fluctuating prevalence of the present taxon in different food types and underscore the importance of continuous monitoring and intervention strategies to mitigate potential risks associated with this pathogen in food safety [25]. Moreover, this study revealed the co-occurrence of biofilm formation and ESBL production in *K. pneumoniae* strains isolated from fresh fruits and vegetables, highlighting the potential risk of these food items as reservoirs of antibiotic-resistant bacteria. Consumption of contaminated fruits and vegetables may lead to the acquisition of antibiotic-resistant pathogens, potentially resulting in foodborne illnesses and treatment challenges. Furthermore, the presence of ESBL-producing *K. pneumoniae* in fresh produce may contribute to the dissemination of resistance genes in the community, exacerbating the global burden of antimicrobial resistance.

In this study, a biofilm detection assay showed that out of the 94 *K. pneumoniae* strains isolated from fresh fruits and vegetables, 76 strains were capable of forming biofilm. This high prevalence of biofilm formation is in agreement with previous studies [26, 27] that reported a strong association between biofilm formation and the ability of bacteria to survive in adverse environments, such as food processing plants and hospital settings. The results showed that out of the 76 biofilm-forming *K. pneumoniae* strains, 74 were ESBL producers and only 2 isolates were ESBL non-producers. ESBL production is a significant mechanism of antibiotic resistance in Gram-negative bacteria, including *K. pneumoniae* [7]. ESBL-producing *K. pneumoniae* strains are increasingly reported worldwide, and their prevalence in food items has become a public health concern [12, 28]. This finding holds significant importance because it highlights the potential correlation between ESBL production and biofilm-forming capabilities in the context of bacterial populations. The identification of a high number of ESBL producers among the biofilm formers suggests a potential association between these two traits, indicating a possible coexistence or linked mechanisms within the bacterial strains [29]. Understanding this association could have implications in society, potentially impacting treatment strategies and infection control measures. The co-occurrence of ESBL production and biofilm formation could influence the resilience of these bacteria and their response to antimicrobial treatments [30], emphasizing the relevance and impact of this specific finding.

ESBL-producing *K. pneumoniae* strains are increasingly reported worldwide, and their prevalence in food items has become a public health concern. The present study is in line with previous studies that reported a high prevalence of ESBL-producing *K. pneumoniae* in various food sources [26, 27]. For instance, a study conducted in Egypt reported the isolation of ESBL-producing *K. pneumoniae* from fresh vegetables [31], while another study conducted in China reported the presence of ESBL-producing *K. pneumoniae* in raw chicken meat [32]. These findings highlight the potential role of food items as reservoirs of antibiotic-resistant bacteria, which can be transmitted to humans through the food chain. The co-occurrence of biofilm formation and ESBL production in *K. pneumoniae* strains isolated from fresh fruits and vegetables is of particular concern, as biofilm formation has been reported to confer increased antibiotic resistance and tolerance to bacteria [33]. Bacterial biofilms are known to protect bacteria from the host immune system and antibiotics, making them difficult to eradicate. Moreover, it has been proven that these pathogens are capable of producing biofilms and are highly resistant to currently available antibiotics. The co-occurrence of biofilm formation and ESBL production in *K. pneumoniae* strains isolated from food items could have serious implications for human health, as these strains can cause severe infections that are difficult to treat [34]. The findings of this study contribute to the existing knowledge regarding the genetic characteristics of ESBL-producing *K. pneum*oniae strains, particularly in the context of food environments. Understanding the genetic profiles and resistance mechanisms of these strains is crucial for developing effective strategies to combat their spread and minimize the risk of infections caused by these MDR pathogens.

Conclusion

The presence of ESBL-producing *K. pneumoniae* in fresh fruits and vegetables presents a critical public health concern. Our observations revealed a noteworthy association between biofilm formation and ESBL production. Notably, the prevalence of ESBLs signifies the importance of this pathogen in fresh produce. This study underscores the imperative need for stringent surveillance and effective control measures to mitigate the risk posed by *K. pneumoniae* in food sources.

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Authors Contributions All authors contributed to the study conception and design. RPS designed, supervised, coordinated the study and drafted the manuscript. SKB and SB, carried out the experiments. KPS performed the sample collection, data analyses and corrected the manuscript. AH analysed the data and edited the manuscript. All authors read and approved the final manuscript.

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Data Availability All data have been submitted in the paper.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical Approval Not applicable.

Consent to Participate Not applicable.

References

1. Vading M, Nauclér P, Kalin M, Giske CG (2018) Invasive infection caused by *Klebsiella pneumoniae* is a disease affecting patients with high comorbidity and associated with high long-term mortality. PLoS ONE 13:e0195258. https://doi.org/10.1371/journal.pone.0195258
2. Castanheira M, Simner PJ, Bradford PA (2021) Extended-spectrum β-lactamases: an update on their characteristics, epidemiology and detection. JAC-Antimicrob Resistance. https://doi.org/10.1093/jacamr/dlab092
3. Rawat D, Nair D (2010) Extended-spectrum β-lactamases in gram negative bacteria. J Glob Infect Dis 2:263–274. https://doi.org/10.4103/0974-777X.68531
4. Tuncer G, Geyiktepe-Guclu C, Surme S et al (2023) Long-term effects of COVID-19 on lungs and the clinical relevance: a 6-month prospective cohort study. Future Microbiol 18:185–198. https://doi.org/10.2217/fmb-2022-0121
5. Wareth G, Neubauer H (2021) The Animal-foods-environment interface of *Klebsiella pneumoniae* in Germany: an observational study on pathogenicity, resistance development and the current situation. Vet Res 52:16. https://doi.org/10.1186/s13567-020-00875-w
6. Junaid K, Ejaz H, Younas S et al (2022) Detection of *Klebsiella pneumoniae* antibiotic-resistant genes: an impending source of multidrug resistance dissemination through raw food. Saudi J Biol Sci 29:3347–3353. https://doi.org/10.1016/j.sjbs.2022.02.020
7. Vatan A, Saltoglu N, Yemisen M et al (2018) Association between biofilm and multi/extensive drug resistance in diabetic foot infection. Int J Clin Pract 72:e13060. https://doi.org/10.1111/ijcp.13060
8. Guerra MES, Destro G, Vieira B et al (2022) *Klebsiella pneumoniae* biofilms and their role in disease pathogenesis. Front Cell Infect Microbiol 12:877995. https://doi.org/10.3389/fcimb.2022.877995
9. Copur B, Dosler S, Aktas Z et al (2022) In vitro activities of antibiotic combinations against mature biofilms of ventilator-associated pneumonia isolates. Future Microbiol 17:1027–1042. https://doi.org/10.2217/fmb-2021-0305
10. Liu X, Yao H, Zhao X, Ge C (2023) Biofilm formation and control of foodborne pathogenic bacteria. Molecules 28:2432. https://doi.org/10.3390/molecules28062432
11. Centeleghe I, Norville P, Hughes L, Maillard J-Y (2023) *Klebsiella pneumoniae* survives on surfaces as a dry biofilm. Am J Infect Control. https://doi.org/10.1016/j.ajic.2023.02.009
12. Giri S, Kudva V, Shetty K, Shetty V (2021) Prevalence and characterization of extended-spectrum β-lactamase-producing antibiotic-resistant *Escherichia coli* and *Klebsiella pneumoniae* in ready-to-eat street foods. Antibiotics 10:850. https://doi.org/10.3390/antibiotics10070850
13. Franklin-Alming FV, Kaspersen H, Hetland MAK et al (2021) Exploring *Klebsiella pneumoniae* in healthy poultry reveals high genetic diversity, good biofilm-forming abilities and higher prevalence in Turkeys than broilers. Front Microbiol. https://doi.org/10.3389/fmicb.2021.725414
14. Sivaraman GK, Sudha S, Muneeb KH et al (2020) Molecular assessment of antimicrobial resistance and virulence in multi drug resistant ESBL-producing *Escherichia coli* and *Klebsiella pneumoniae* from food fishes, Assam. India Microbial Pathog 149:104581. https://doi.org/10.1016/j.micpath.2020.104581
15. Denissen J, Reyneke B, Waso-Reyneke M et al (2022) Prevalence of ESKAPE pathogens in the environment: antibiotic resistance status, community-acquired infection and risk to human health. Int J Hyg Environ Health 244:114006. https://doi.org/10.1016/j.ijheh.2022.114006
16. Prasad M, Shetty SK, Nair BG et al (2022) A novel and improved selective media for the isolation and enumeration of *Klebsiella* species. Appl Microbiol Biotechnol 106:8273–8284. https://doi.org/10.1007/s00253-022-12270-w
17. Banerjee S, Bajire SK, Mithun HK, Shastry RP (2023) 3-(Bromoacetyl) coumarin is a potential therapeutic agent against neonatal sepsis-associated *Pseudomonas extremorientalis*. Arch Microbiol 205:312. https://doi.org/10.1007/s00203-023-03653-2
18. CLSI (2020) Performance standards for antimicrobial susceptibility testing, twenty-fifth informational supplement in M100–S30. Clinical and Laboratory Standards Institute, Wayne, PA
19. Shastry RP, Ghate SD, Banerjee S (2021) Culture dependent and independent detection of multiple extended beta-lactamase producing and biofilm forming *Salmonella* species from leafy vegetables. Biocatal Agric Biotechnol 38:102202. https://doi.org/10.1016/j.bcab.2021.102202
20. Neopane P, Nepal HP, Shrestha R et al (2018) In vitro biofilm formation by *Staphylococcus aureus* isolated from wounds of hospital-admitted patients and their association with antimicrobial resistance. Int J Gen Med 11:25–32. https://doi.org/10.2147/IJGM.S153268
21. Stepanović S, Vuković D, Hola V et al (2007) Quantification of biofilm in microtiter plates: overview of testing conditions and practical recommendations for assessment of biofilm production by *staphylococci*. APMIS 115:891–899. https://doi.org/10.1111/j.1600-0463.2007.apm_630.x
22. Bajire SK, Jain S, Johnson RP, Shastry RP (2021) 6-Methylcoumarin attenuates quorum sensing and biofilm formation in *Pseudomonas aeruginosa* PAO1 and its applications on solid surface coatings with polyurethane. Appl Microbiol Biotechnol. https://doi.org/10.1007/s00253-021-11637-9
23. Grover N, Sahni AK, Bhattacharya S (2013) Therapeutic challenges of ESBLS and AmpC beta-lactamase producers in a tertiary care center. Med J Armed Forces India 69:4–10. https://doi.org/10.1016/j.mjafi.2012.02.001
24. Ben Said L, Jouini A, Klibi N et al (2015) Detection of extended-spectrum beta-lactamase (ESBL)-producing *Enterobacteriaceae* in vegetables, soil and water of the farm environment in Tunisia. Int J Food Microbiol 203:86–92. https://doi.org/10.1016/j.ijfoodmicro.2015.02.023

25. Sheng L, Wang L (2021) The microbial safety of fish and fish products: recent advances in understanding its significance, contamination sources, and control strategies. Compr Rev Food Sci Food Saf 20:738–786. https://doi.org/10.1111/1541-4337.12671
26. Shadkam S, Goli HR, Mirzaei B et al (2021) Correlation between antimicrobial resistance and biofilm formation capability among *Klebsiella pneumoniae* strains isolated from hospitalized patients in Iran. Ann Clin Microbiol Antimicrob 20:13. https://doi.org/10.1186/s12941-021-00418-x
27. Seifi K, Kazemian H, Heidari H et al (2016) Evaluation of biofilm formation among *Klebsiella pneumoniae* isolates and molecular characterization by ERIC-PCR. Jundishapur J Microbiol 9:e30682. https://doi.org/10.5812/jjm.30682
28. Ugbo EN, Anyamene CO, Moses IB et al (2020) Prevalence of blaTEM, blaSHV, and blaCTX-M genes among extended spectrum beta-lactamase-producing *Escherichia coli* and *Klebsiella pneumoniae* of clinical origin. Gene Rep 21:100909. https://doi.org/10.1016/j.genrep.2020.100909
29. Amanatidou E, Matthews AC, Kuhlicke U et al (2019) Biofilms facilitate cheating and social exploitation of β-lactam resistance in *Escherichia coli*. npj Biofilms Microbiomes. https://doi.org/10.1038/s41522-019-0109-2
30. Dutt Y, Dhiman R, Singh T et al (2022) The association between biofilm formation and antimicrobial resistance with possible ingenious bio-remedial approaches. Antibiotics 11:930. https://doi.org/10.3390/antibiotics11070930
31. Elshafiee EA, Kadry M, Nader SM, Ahmed ZS (2022) Extended-spectrum-beta-lactamases and carbapenemase-producing *Klebsiella pneumoniae* isolated from fresh produce farms in different governorates of Egypt. Vet World 15:1191–1196. https://doi.org/10.14202/vetworld.2022.1191-1196
32. Li Z, Xin L, Peng C et al (2022) Prevalence and antimicrobial susceptibility profiles of ESBL-producing *Klebsiella Pneumoniae* from broiler chicken farms in Shandong Province. China Poult Sci 101:102002. https://doi.org/10.1016/j.psj.2022.102002
33. Romyasamit C, Sornsenee P, Chimplee S et al (2021) Prevalence and characterization of extended-spectrum β-lactamase-producing *Escherichia coli* and *Klebsiella pneumoniae* isolated from raw vegetables retailed in Southern Thailand. PeerJ 9:e11787. https://doi.org/10.7717/peerj.11787
34. Vuotto C, Longo F, Balice MP et al (2014) Antibiotic resistance related to biofilm formation in *Klebsiella pneumoniae*. Pathogens 3:743–758. https://doi.org/10.3390/pathogens3030743
35. Malik VK, Singh M, Sangwan P et al (2023) First report of *Klebsiella* leaf streak on *Sorghum* caused by *Klebsiella variicola* in Haryana India. Plant Dis. https://doi.org/10.1094/PDIS-09-22-2200-PDN
36. Chanseyha C, Bilal Sadiq M, Zar Aye Cho T, Kumar Anal A (2018) Prevalence and analysis of antibiotic resistant genes in *Escherichia coli* and *Salmonella* isolates from green leaf Lettuce—Chiang Mai Journal of Science. Chiang Mai J Sci 45:1274–1286
37. Raseala CM, Ekwanzala MD, Momba MNB (2020) Shared extended-spectrum β-lactamase-producing *Salmonella* serovars between agricultural and aquatic environments revealed through invA amplicon sequencing. Microorganisms 8:1898. https://doi.org/10.3390/microorganisms8121898
38. Shahid M, Malik A, Adil M et al (2009) Comparison of beta-lactamase genes in clinical and food bacterial isolates in India. J Infect Dev Ctries 3:593–598. https://doi.org/10.3855/jidc.550
39. Njage PMK, Buys EM (2015) Pathogenic and commensal *Escherichia coli* from irrigation water show potential in transmission of extended spectrum and AmpC β-lactamases determinants to isolates from lettuce. Microb Biotechnol 8:462–473. https://doi.org/10.1111/1751-7915.12234